

FIND: Flexible Intelligence through Neurocomputational Design

In the next decade, we envision AI making the leap to human-like cognitive flexibility—requiring lasting memory, causal understanding, and continuous adaptation. Achieving this demands a paradigm shift in how neural networks are built, drawing inspiration from biological memory systems.

1 Excellence

We live in time. Memory serves as the essential bridge connecting our past to our present, enabling an agent to maintain continuity of experience, learn from previous encounters, and make informed decisions. Humans can develop internal models of how the world works, enabling planning and reasoning by simulating potential outcomes. Replicating this ability in machines might be the next step toward flexible AI agents [1]. Compared to humans, current AI systems struggle to generalize their knowledge to new situations [2]. AI is limited by the lack of long-term memory [3], causal understanding [4], and the ability to adapt through life-long learning without catastrophic forgetting [5]. The prevailing technologies—recurrent neural networks, state space models, and attention-based models—remain constrained by their bounded computational frameworks [3, 6]. In scenarios where it is necessary to understand the effects of past events, stable long-term memory is essential. This raises the question: **How can systems intelligently select what information to retain or discard, and what to retrieve, in order to learn across long temporal sequences?** In FIND, we aim to address this challenge through three work packages, studying how modularized memory-augmented neural networks:

- **WP1:** Process symbolic information over different time scales
- **WP2:** Represent cause and effect of actions in world models
- **WP3:** Continuously adapt to changing environments

We propose the development of neurocomputational networks where modules of neurons are capable of reading and writing to local long-term memory. This novel framework, **PAM (Predictive Active temporal coding with distributed Memory)**, adheres to the “shallow brain hypothesis” [7] and operates through:

- Specialized neural modules with dedicated memory systems [8, 9, 10]
- Symbolic representations [11] retained in memory
- Learned orchestration mechanisms that dynamically route information based on task demands
- Probabilistic processing of inputs and actions to generate temporal predictions
- Action selection that minimizes uncertainty [12, 13, 14, 15]

FIND’s long-term goal is to mimic the brain’s memory capacity for exploration, reasoning, and adaptive knowledge transfer. We will test PAM in game environments like mazes, Minecraft, and Factorio, providing insights into efficient and adaptable networks for complex planning, resource management, and causal reasoning.

1.1 State of the art, knowledge needs and project objectives

Long-term memory

A key challenge in AI is creating models with similar memory capabilities to the brain. Recent advances have unified in-context learning with both retrieval-augmented generation (RAG) [16, 17] and reinforcement learning (RL), creating more adaptive AI systems. While RAG enhances in-context learning by dynamically retrieving relevant information, in-context learning has also impacted RL by enabling few-shot policy learning from demonstration trajectories [18]. However, these enhancements ultimately serve as palliative solutions that sidestep the fundamental challenge of bottom-up lifelong learning, as they remain dependent on human-curated memory augmentation and struggle with out-of-distribution scenarios [2]. **These failures highlight the gap between current AI**

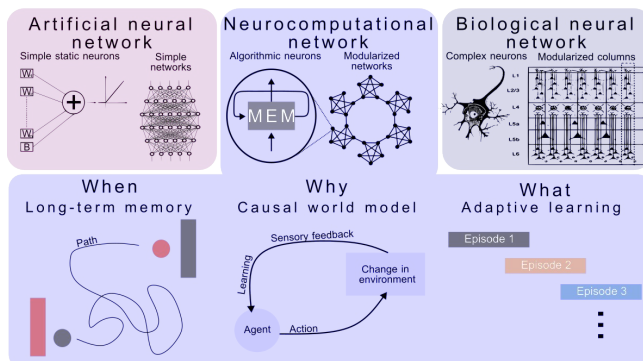


Figure 1: Conceptual overview of FIND. We develop neurocomputational networks whose modules hold local long-term memories, striking an algorithmic balance between overly simple ANNs and overly complex biological neural networks. Computations are orchestrated across specialised modules to address three core challenges: *long-term memory* (recalling past experiences during decision making); *causal reasoning* (learning cause-effect relationships through active environmental interventions); and *adaptive learning* (acquiring new knowledge continuously without forgetting).

systems and truly autonomous learning agents that can continuously adapt and generalize through direct experience. One striking example: Claude 3.7 Sonnet spent over 80 hours wandering in circles in Pokémon Red, unable to recall where it had already been [19]—a symptom of the fundamental memory limitation FIND addresses. **Memory is essential in this context, for tasks like planning, reasoning, and learning, which require the ability to store and recall information efficiently.**

The core challenge for long-term memory in neural networks is separating memory storage from computation [20, 21]. Memory-augmented neural networks (MANNs) have been shown useful to mitigate this issue. The success of MANNs depends on effective memory access mechanisms, as well as balancing the trade-off between fast adaptation and preservation of existing information. MANNs include sophisticated content-based addressing schemes [22, 23]. Bayesian models for online memory compression and generative approaches have also been explored, as seen in the Kanerva Machines [8, 10, 24, 25], drawing inspiration from sparse distributed memories (SDM) [9]. Product Kanerva Machines [10] use a hierarchical structure to combine smaller Kanerva Machines, abstracting invariant features and improving scaling and capacity. Leading perspectives in the field hold that human-level reasoning requires integrating symbolic AI with neural networks [26]. Hyperdimensional computing extends SDM by enabling symbolic operations through vector binding and composition, offering a promising approach to integrate neural and symbolic reasoning capabilities within a unified framework [11, 27]. **Despite these advances, significant challenges remain in developing memory systems that effectively bridge neural and symbolic representations while compressing temporal information to support both statistical learning and transparent and explicit reasoning.** FIND adopts a neurosymbolic approach, leveraging hyperdimensional computing [11] and recent advances [27, 28] to enable explicit, compositional reasoning alongside statistical learning.

Causal world model

A causal world model represents not just statistical associations between states but the generative mechanisms by which actions produce effects—enabling counterfactual reasoning and robust generalisation to novel situations [29, 4]. Much like the human brain constructs mental models to navigate reality, AI systems require sophisticated world models to achieve flexible intelligence. Recent developments in this field include Joint-Embedding Predictive Architectures (JEPA) [30], which learn representations through prediction tasks, enabling a more robust understanding of environmental dynamics. Similarly, DINO-WM [31] has demonstrated capability in modeling world dynamics and performing task-agnostic planning, representing a significant step toward more flexible AI systems. Using Structured State Space Sequence (S4) in world models (S4WM) has shown promise in long-term memory problems in model-based reinforcement learning [6]. However, the time horizons typically used in these

benchmarks are irrelevant for real-world problems. For example, using 2000 steps as benchmarks [6] for long-term memory does not even reflect the capabilities of desert ants, which hold accurate memories after 30–200 000 steps [32]. When considering the vast memory capacity required for a full lifetime of learning, models like transformers and S4 face critical limitations in long-term memory and credit assignment due to their limited context windows [33], highlighting the need for further research in this area.

The development of effective AI systems requires understanding both extrinsic motivation (external rewards) and intrinsic motivation (internal drives for exploration and novelty) while capturing long-term dependencies, handling environmental uncertainty, and maintaining computational efficiency. Latent learning of cognitive maps has been pursued in neuroscience [34, 35] and AI [36, 37], including by our lab [38, 39, 40, 41], but these approaches lack a unified theory of perception, cognition, and action. While these spatial representations encode where an agent has been, they lack an explicit mechanism for what caused transitions—the gap FIND addresses through memory-augmented temporal predictive coding. According to Bayesian theories, the brain interprets sensory data through probabilistic inference of latent causes. Predictive coding (PC) offers a unified approach by minimizing prediction errors between top-down expectations and bottom-up inputs, discovering novel information, and combining internal goals with external rewards [13, 12]. This framework has recently been extended to active and temporal PCs, allowing the system to actively sample its environment to minimize uncertainty [12, 15] and enabling biologically plausible cognitive maps and prediction of sensory input sequences [14, 42, 43, 44, 45]. **However, these models rely on Markov assumptions, where predictions depend only on the previous state, making them unable to efficiently handle long-term memory and credit assignment or prevent catastrophic forgetting.** FIND addresses these limitations by implementing modular memory-augmented neural networks that transcend Markov assumptions through distributed temporal processing at multiple

time scales, combining predictive coding with hyperdimensional computing to enable both long-term memory retention and causal understanding while preventing catastrophic forgetting.

Adaptive learning

Adaptable memory for life-long learning requires systems to dynamically manage information in open-ended environments without catastrophic forgetting, balancing stability with plasticity. Current approaches span regularization, replay buffers, and Bayesian variational methods [5, 46, 47, 48]; in our own lab we have explored generative approaches to both stability [49] and plasticity [50]. While these frameworks provide principled foundations, they face a fundamental bottleneck: as experiences accumulate, intelligent content-based selection of *what* to store and *what* to retrieve becomes critical—a challenge statistical methods alone cannot solve. FIND addresses this through hyperdimensional computing [11, 27]: by binding memories as compositional high-dimensional vectors, the system enables both fast approximate retrieval (nearest-neighbour in HDC space) and explicit symbolic queries (binding and unbinding operations), supporting life-long learning without retraining from scratch. Life-long learning additionally requires hierarchical abstractions—identifying patterns at multiple scales from immediate object interactions to long-term causal chains—which FIND achieves through its multi-module architecture with dedicated slow and fast memory systems.

In a modularized system, different neural modules can specialize in processing information at distinct abstraction levels, from sensory details to high-level causal relationships, with each maintaining optimized memory representations for its particular scale. The orchestration mechanisms that coordinate module communication create pathways between these abstraction levels, enabling efficient compositional reasoning where complex ideas can be built from simpler components while facilitating knowledge transfer across domains. **FIND’s integrative approach, combining biological insights with computational innovation through our proposed framework, PAM, provides a concrete path toward AI systems capable of sophisticated reasoning, efficient learning, and adaptive behavior.**

1.2 Research questions and hypotheses, theoretical approach, and methodology

The BioAI group at Simula has developed the foundations of FIND through a coherent line of research on neurocomputational memory and orchestration. In computational neuroscience, we have established that AI agents spontaneously develop grid, place, and border cells [38], and that orthogonal remapping of representational geometry encodes context-dependent memory [40, 41, 39]. In neural orchestration, we have demonstrated that Neural Cellular Automata (NCA) can serve as self-organizing inter-module communication mechanisms [51], that NCA-based engram networks support one-shot memory transfer [52], and that NCA produce developmental solutions to abstract reasoning tasks [53]. Building directly on these results, we have implemented and benchmarked the outline of the PAM architecture: the tPC-KM system achieves 98.1% exact recall at 40% input noise ($\sigma=0.4$, $K=100$ items, MNIST) versus 56.1% for a feedforward encoder with identical memory—a $1.75\times$ gap attributable solely to iterative inference. Under online concept drift ($N_{\text{drift}}=200$ items), the system forgets $3.75\times$ less than the tPC-alone baseline (forgetting 0.008 vs. 0.030). A 200-step temporal persistence benchmark further confirms that memory is retained across extended sequences, with a manuscript in preparation. These results confirm the feasibility of the proposed approach and directly motivate extending to harder, naturalistic datasets in T1.1.

Main hypothesis: Modular specialization can emerge naturally from predictive principles and provide long-term memory support for active inference and adaptive learning, enabling causal world models without catastrophic forgetting. We will evaluate this hypothesis by answering the following **research questions:**

- Q1: How can memory-augmented modules distribute and combine long-term temporal information?
- Q2: How can such modules support active inference?
- Q3: How can neurosymbolic components be integrated into distributed memory architectures without sacrificing gradient-based learning?
- Q4: Does this integration measurably enhance compositional reasoning in complex environments?
- Q5: Can such networks learn adaptively and reason about causal relations?

These research questions will be addressed in the following **work packages:**

- WP1: Temporal Predictive Coding with Distributed Memory (Q1)
- WP2: Active Temporal Predictive Coding with Distributed Memory (Q2–Q3)
- WP3: Adaptive Causal Learning and Reasoning in Open-Ended Environments (Q4–Q5)

Q1 is addressed in WP1; Q2–Q3 in WP2; Q4–Q5 in WP3. The WPs build sequentially: PM (WP1) provides the passive memory foundation, PAM (WP2) adds active inference and neurosymbolic integration, and WP3 stress-tests the complete system in open-ended environments.

Theoretical approach: PAM combines three key mechanisms into a unified architecture: (1) *Distributed modular specialization*: Instead of having a single centralized memory system, PAM distributes memory across multiple specialized modules. Each module has its dedicated memory space, inspired by Kanerva machines, and can process information independently. (2) *Temporal predictive coding*: The system constantly generates predictions about future sensory inputs based on its current state. When there is a mismatch between prediction and actual input (prediction error), the system updates its internal model, including memory, to improve future predictions. (3) *Active inference*: Rather than passively observing, the system actively chooses actions to minimize uncertainty about its predictions. This allows it to explore and learn about its environment efficiently and store causally relevant memories.

Our approach introduces a novel framework that fundamentally reimagines predictive coding through a modular architecture with distributed memories. Rather than relying on traditional hierarchical structures, we propose a network of specialized processing units with memory systems guiding temporal predictions that interact through learned orchestration mechanisms. Memory modules communicate through learned network structures, allowing information to flow dynamically between specialized processors operating at different time scales. Distributing memories reduces computational load [10] while specialized modules can factorize conditional distributions, modeling the causal structure of the data-generating process [29, 4]. Directing processes according to prediction errors, based on Bayesian theory, balances plasticity when focusing on the problem at hand with stability of important previous knowledge [8, 46, 47].

One of the challenges in designing effective modularized AI ecosystems is balancing redundancy and enabling functional specialization. FIND builds on ongoing work in our lab [54] and takes advantage of biological constraints, such as non-negativity and energy efficiency, which can promote disentangled and robust representations [55], ensuring modules focus on distinct factors of variation. Furthermore, self-supervised learning of contextual separation of cognitive maps [39, 38, 40] will be essential for establishing distinct reference frames [56] and enabling the identification of modular structures in task domains.

To make the approach precise, we give the core update equations that underpin PAM’s three mechanisms. Each module, indexed by m , has a latent state $z_t^m \in \mathbb{R}^{d_z}$ and sensory input $x_t^m \in \mathbb{R}^{d_x}$ at time t . The prediction error $e_t = x_t^m - \hat{x}_t^m$ is calculated from a sensory prediction obtained from the latent state. Module state updates according to $z_{t+1}^m = f_m(z_t^m, c_t^m, r_t^m, a_t)$, where f_m is a learnable function. The update is driven by action a_t , context-dependent memory $r_c^m = \sum_i \alpha_c^m[i] h_i^m$, with weight α , and contextual input c_t^m . The context is determined by one of the following orchestration mechanisms we propose to explore:

1. **Message passing (primary)**: message transformations, such as attention and convolution, combined with aggregation functions—our lab has demonstrated effective emergent module communication with this mechanism in active vision tasks [51]
2. **Neural cellular automata (NCA, primary)**: [54] self-organizing principles determine module communication—demonstrated in one-shot memory transfer [52] and abstract reasoning [53]

Four further mechanisms form a structured ablation ladder to be evaluated in WP2, enabling us to quantify each design choice’s contribution independently: *Hypernetwork* (auxiliary network generates inter-module parameters), *Predictive routing* (modules proactively route information to anticipated recipients), *Competitive collaboration* (modules compete for influence but must collaborate on tasks), and *Energy-based orchestration* (information flow governed by an energy landscape with thermodynamic-like phase transitions). Message passing and NCA are designated primary because they are the only mechanisms for which our lab already has working, benchmarked implementations—message passing demonstrated in active vision [51] and NCA in one-shot memory transfer [52] and abstract reasoning [53]—providing a calibrated baseline from which the WP2 ablation ladder will proceed in order of increasing biological plausibility and implementation complexity.

PAM is based on content-addressable memory [8, 23] which relies on a similarity score between the current latent state $z_t^m \in \mathbb{R}^{d_m}$ and the stored memory basis vectors $h_i^m \in \mathbb{R}^{d_h}$ to determine which memories to retrieve or update. The precise form of the retrieval and update rule can vary based on the orchestration mechanism, as a starting point we will continue on the current work in our lab that uses exponential maps [57], which respect the curved geometry of high-dimensional hypervectors and enable closed-form Riemannian updates without re-normalisation—implemented as a single differentiable module. This approach implements storage and forgetting through posterior

updating, where memories are preserved based on predictive value while redundant information is consolidated and balanced through Bayesian theory [8, 46] and complementary learning systems [24].

PAM leverages hyperdimensional computing [11, 57] to bridge neural and symbolic approaches in two critical ways: (1) implementing sparse distributed memories where symbolic relationships are encoded as high-dimensional vectors, and (2) enabling orchestration mechanisms to perform binding operations for compositional reasoning across modules. This unified approach supports both statistical learning and symbolic manipulation, allowing PAM to perform explicit reasoning while maintaining the efficiency of distributed neural representations. For example, a sequence of object encounters can be encoded as a superposition of bound (object–location) pairs; querying with a partial key retrieves the most similar stored pattern in $O(d)$ operations regardless of memory size [11]. The integration of actions is formalized to process sensory inputs and actions through a probabilistic framework, where actions serve as both predicted outcomes and causes of state transitions. Actions are selected to minimize uncertainty about predictions influenced by individual modules.

WP1 Temporal Predictive Coding with Distributed Memory (PM) (Lepperød, PD)

This WP aims to establish the core architecture and mechanisms for temporal prediction and memory within a passive observation framework. Despite the preliminary work presented, there are core architectural and mechanistic elements that need to be established for temporal prediction and memory within a passive observation framework. **Objectives:** (1) Develop the foundational PM architecture with integrated memory modules. (2) Implement and evaluate multi-scale temporal prediction capabilities. (3) Analyze memory encoding, retrieval, and probing techniques in passive tasks to understand learned representations.

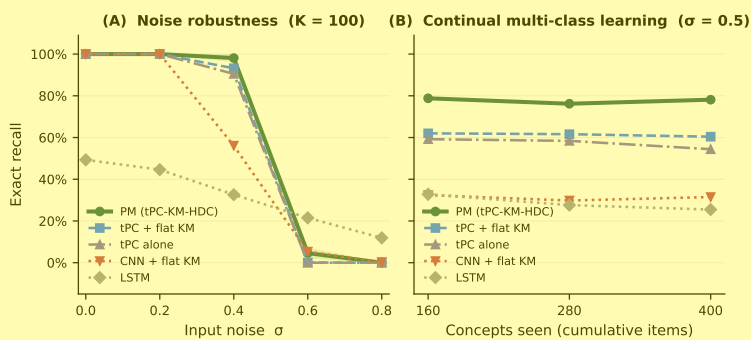


Figure 2: Preliminary PM results (WP1 foundation). (A) Exact recall vs. input noise σ ($K=100$ items, 20 trials). All four memory-augmented models achieve near-perfect recall at $\sigma \leq 0.2$; at $\sigma=0.4$ iterative predictive inference (PM, tPC-KM) outperforms feedforward encoding (CNN-KM) by a $1.75\times$ margin; LSTM is noise-floor limited throughout. (B) Exact recall across three sequential class phases ($\sigma=0.5$, cumulative items 160/280/400). PM maintains $\approx 78\%$ recall across all phases; tPC alone drops 4.8 pp by phase 3, and LSTM drops 7.3 pp, confirming that HDC binding prevents inter-class interference as new concepts accumulate. Shading (A) shows ± 1 s.d. across trials.

volutional techniques such as ResNet [61]. We will compare PM against existing GTMs [20]—VRNNs [62], LSTMs [63], and S4 [64]—and memory-augmented neural networks—NTMs [22] and DNCs [23]. Offline probing techniques [65] will be used to analyze representations learned by PM, training separate probe networks to predict relevant properties of the input sequence. *Quantitative success criterion:* PM must achieve $>90\%$ exact recall at $\sigma=0.3$ and maintain above-chance performance across all benchmark phases.

Task 1.2: Inference Depth, Noise Robustness, and Capacity Characterisation

Preliminary results reveal that PM’s advantage over CNN-KM and tPC-alone depends critically on iterative predictive-coding inference: removing it degrades recall by $1.75\times$ at $\sigma=0.4$. However, fixed-iteration inference is computationally wasteful for easy inputs. This task will (i) systematically characterise the full recall–noise–capacity landscape across $\sigma \in [0.0, 0.8]$ and $K \in [10, 500]$ to identify the operational regime of PM; (ii) implement adaptive inference depth via prediction-error thresholding; and (iii) characterise the interaction between slot count, HDC dimension, and forgetting rate under varying concept-drift speeds, providing design guidelines for WP2–3. *Quantitative success criterion:*

Preliminary results. Figure 2 summarises two independent benchmarks. Panel A shows that iterative predictive-coding inference lifts exact recall from 56% to 93–98% at $\sigma=0.4$ relative to a feedforward encoder with identical memory; LSTM is limited to $\sim 49\%$ throughout. Panel B shows that HDC-based binding reduces inter-class forgetting $3.75\times$ versus tPC alone across three sequential class phases. These results motivate T1.1–T1.3.

Task 1.1: Benchmarks on Passive Temporal Prediction Tasks

Completed: MNIST digit recall [20] benchmark is complete; results are summarised in the preliminary results above. **Planned:** Omniglot [58], the Malmo environment in Minecraft [59], “infinite dSprites” [60], and Memory Maze offline probing benchmarks [3].

Visual input will be encoded with con-

Adaptive inference must achieve $\geq 95\%$ of full-iteration accuracy at $\leq 60\%$ of compute on the MNIST benchmark.

Task 1.3: Hierarchical Temporal Compression for Long-Horizon Memory

Current benchmarks confirm 200-step temporal persistence, but cognitively relevant tasks require memory across thousands to millions of timesteps (desert ants maintain accurate path integration beyond 200,000 steps [32]; Factorio production chains span $>10,000$ decisions). This task extends PM with hierarchical time-scale factorisation: fast modules ($\Delta t=1$) process step-level inputs, while slow modules compress episodes into semantic summaries via a learned gating mechanism inspired by complementary learning systems [24]. The gating criterion—whether and when prediction error should trigger consolidation from episodic to semantic memory—is an open design question; the threshold and window size will be resolved experimentally in the first months of T1.3, using prediction-error statistics from T1.2 as initial guidance. We will evaluate on “infinite dSprites” [60] and 2,000–10,000 step sequences in the Memory Maze [3], with long-horizon generalisation as the primary success criterion.

WP2 Predictive Active Temporal Coding with Distributed Memory (PAM) (Lepperød, PD)

This WP extends PM by incorporating active inference capabilities. The research integrates action selection, planning, and causal reasoning while maintaining efficient memory utilization. Building on demonstrated NCA-based active vision [51] and one-shot engram transfer [52], WP2 extends these mechanisms to action-conditioned temporal prediction, motivated by recent findings on neurosymbolic integration [28], enhancing the system’s ability to model causal dependencies and dynamically adapt to changing environments. **Objectives:** (1) Develop an active predictive coding framework integrating action selection with distributed memory. (2) Implement and evaluate prediction-driven action selection capabilities. (3) Analyze how actions influence memory encoding and retrieval. (4) Study action-conditioned temporal predictions.

Task 2.1: Action-Augmented Temporal Predictive Coding

Extend the PM architecture to incorporate action nodes that modulate state transitions, developing methods for action-dependent prediction error computation across multiple timescales. Design and implement mechanisms for integrating actions into the distributed memory framework, including action-conditioned memory addressing, action-dependent update rules for memory contents, and methods for maintaining temporal consistency in action-state representations. Develop comprehensive learning algorithms that jointly optimize state predictions and action selections, incorporating both local and global learning rules. *Quantitative success criterion:* Action-conditioned prediction error must correlate $r > 0.7$ with observed state transitions in a controlled navigation task.

Task 2.2: Active Memory Access and Update Mechanisms

Create a complete system for action-influenced memory operations including content-addressable retrieval, priority-based updating, and temporal coherence maintenance. Standard content-based addressing (attention), residual connections between hierarchical levels [15], and experience replay based on action outcomes [66] form the engineering substrate; the research contribution is the action-driven consolidation rule that decides when to update, prune, or reinforce memory slots based on prediction-error signals. Develop pruning strategies that maintain task-relevant information under memory constraints. *Quantitative success criterion:* Active memory consolidation must reduce memory footprint by $\geq 30\%$ with $< 5\%$ recall degradation on WP1 benchmarks.

Task 2.3: Action Selection through Predictive Processing

Implement an active inference framework for action selection that accounts for both information gain and goal-oriented reward, incorporating uncertainty estimation and risk assessment. Develop an action representation scheme that enables both abstract policy learning and detailed motor control, including methods for learning action primitives and combining them into complex sequences, based on our work [54] and others [15]. Create adaptive exploration strategies balancing information gain and task performance. Implement methods for exploring large action spaces and develop criteria for switching between exploration and exploitation. Implement proximal policy optimization (PPO) [67] for credit assignment-based tasks. Explore pretraining using self-supervised exploration combined with imitation learning [68]. *Quantitative success criterion:* PPO with active-inference exploration must reach task completion $2\times$ faster than standard PPO on at least two benchmark environments.

Task 2.4: Evaluation using Active Learning Tasks

Evaluate PAM against benchmarks for understanding the system: environments include few-shot benchmarks in continuous control tasks [68, 69, 70], and long-horizon planning problems [3], analyzing learning curves and computational requirements. Execute comprehensive ablation studies to

understand the contribution of each component (orchestration mechanisms, distributed memory, action selection mechanisms) to overall performance. Investigate how the computations performed by the PM model can be mapped onto neural circuits by extending our established RNN framework [38, 39, 40] that demonstrates the spontaneous development of grid, place, and border cells. The analysis will include population-level activity in grid cells' population structure [41] and single-cell interactions through ablation studies. *Quantitative success criteria:* PAM must achieve a statistically significant improvement (>10 percentage points) over the LSTM baseline on Memory Maze long-horizon planning ($T > 500$ steps), and reduce forgetting below the tPC-alone threshold (forgetting < 0.015) under active concept drift with $N_{\text{drift}} = 200$. Orchestration ablations must show that the primary NCA/message-passing design outperforms at least two alternative mechanisms on combined noise-robustness and forgetting metrics.

WP3: Adaptive Learning and Causal Reasoning in Open-Ended Environments (Lepperød, PD)

This WP will put the system to the test in environments that demand sophisticated planning, resource management, and adaptation to ever-changing dynamics. We will evaluate adaptive learning and causal reasoning abilities of PAM in complex, dynamic environments, specifically Minecraft [71] and Factorio [72]. These games serve as robust cognitive benchmarks through distinct gameplay mechanics. Factorio tests long-term memory through complex production chains and factory layouts, while Minecraft challenges spatial recall across vast procedurally generated worlds. For causal reasoning, Factorio requires understanding intricate resource flows and production bottlenecks, whereas Minecraft demands mastery of redstone logic circuits and environmental interactions. Adaptive learning manifests differently in each: Factorio emphasizes iterative optimization of industrial systems, while Minecraft rewards creative problem-solving in unpredictable environments. PAM will address these challenges through its modular architecture, action-conditioned memory, and hierarchical control. **Objectives:** (1) Investigate how PAM learns causal relationships through active inference, particularly in open-ended environments. (2) Evaluate the system's support of curriculum and adaptive learning. (3) Examine how the modular structure of PAM facilitates knowledge transfer. (4) Analyze task and action space representations learned by the system.

Task 3.1: Environment Development and Curriculum Design

Design complex, dynamic scenarios in Minecraft and Factorio presenting varied challenges, including resource management, crafting, building, combat, and exploration. Create a procedural generation pipeline that systematically varies task complexity, resource constraints and causal dependencies. Share and open source all test cases and data. Implement a learning progress-based curriculum driven by metrics such as task completion time, resource efficiency, and the complexity of actions performed. Adjust difficulty level dynamically based on the agent's performance, using automatic curriculum learning (ACL) [73] and reinforcement learning with intrinsic motivation [74]. Implement a task-switching mechanism that allows explicit task signals to control the current objective for controlled experiments. Integration will use the existing Python Malmo API [59] for Minecraft and the community Factorio RCON interface; the observation vector will comprise visual frames (encoded via a frozen ResNet encoder [61]) and game-state variables. A simplified wrapper environment will be developed and validated in the first month of WP3 before full PAM integration begins.

Task 3.2: Active Inference Implementation

Integrate the PAM architecture into an interactive agent capable of operating within Minecraft and Factorio, handling diverse inputs (visual, game state) and outputs (movement, item interactions, building actions). We implement a three-level policy hierarchy: (L1) step-level motor primitives learned by the lowest PAM modules via PPO [67]; (L2) sub-goal selection learned by mid-level modules via hindsight experience replay [75]; (L3) strategic planning by the top modules, represented as symbolic goal vectors in HDC space [11, 28]. This hierarchy will be co-trained through a combination of reinforcement learning and self-supervised prediction objectives.

Task 3.3: Causal and Adaptive Learning Analysis

Assess causal relationship learning using specific methods like interventions (e.g., changing game rules or introducing disturbances), counterfactual reasoning (e.g., simulating alternative scenarios), and analyzing the agent's ability to predict the effects of its actions. We expect to see specialization based on temporal scale (e.g., short-term vs. long-term planning), task domain (e.g., combat vs. building), and abstraction level (e.g., high-level goals vs. low-level actions). Specialization will be measured by analyzing the activity patterns of individual modules, the types of information stored in their memories, and their contribution to different tasks. *Quantitative success criterion:* Intervention

experiments must demonstrate >20% accuracy in predicting the effect of a novel action intervention not seen during training.

Task 3.4: Environmental Adaptation

Compare performance against RL baselines: DQN [76], PPO, and RND [77]. Perform systematic comparison with baseline world model algorithms (Dreamer [78], SEED RL [79], DINO-WM [31], MuZero [80]), analyzing sample efficiency, computational cost, and generalization capabilities. Analyze memory organization and retrieval patterns, looking for temporal sequences, action-outcome associations, and hierarchical organization through visualization of memory contents and analysis of retrieval patterns during different tasks. *Quantitative success criteria:* (i) PAM must achieve >10 pp improvement over DQN on at least two Minecraft sub-tasks after 10⁶ environment steps. (ii) In Factorio, PAM must reach a production throughput (items/min) at least 1.5× that of a DreamerV3 baseline trained for the same number of steps on the automation-tier task, demonstrating that memory-augmented causal reasoning confers an advantage on long-horizon resource planning beyond what world models without explicit memory achieve.

Risks and Mitigation

Category	Description	Like.	Sev.	Mitigation
Scientific	Modular architecture may fail to effectively specialize.	Med	High	Progressive training, soft inductive biases, explicit architectural constraints.
	Distributed memory may not overcome catastrophic forgetting.	Med–High	High	Hybrid memory combining fast and slow learning techniques.
	Memory-augmented tPC failing to scale to complex environments.	Med	Med	Start with simplified environments; adaptive memory pruning.
Technical	Integration complexity between components.	High	High	Modular software design with well-defined interfaces.
	Computational resource requirements exceed expectations.	Med	Med	Optimized distributed computing; model distillation; adaptive memory compression.
	Orchestration bottlenecks in information flow.	Med–High	High	Benchmark orchestration designs with sparse activation patterns.
	Integration of multiple learning mechanisms may lead to instability.	High	Med	Careful gradient monitoring; progressive training schedules.
Impl.	Environment integration (Minecraft/Factorio) challenging.	Med	Med	Develop simplified environment wrappers with fallback options.
	Debugging complexity in distributed systems.	High	High	Comprehensive visualization and component-testing tools.
Ethical	High energy consumption.	Med–High	Med	Prioritize energy efficiency in system design.
	Limited explainability and reproducibility.	Med	High	Memory access visualization, interpretability tools, rigorous version control.
Project	Interdependencies causing cascading delays.	Med	High	Agile management practices; identify critical paths.
	Difficulty recruiting specialized expertise.	Med	Med	Leverage collaboration networks and external expertise.
Project mgmt.	Failure to reach project milestones.	Med	Med	Quarterly progress evaluations to enable strategic adjustments.

Gender, security, and ethical considerations

FIND will comply with the guidelines issued by Norway’s National Committee for Research Ethics in Science and Technology and follow the Vancouver-based publication guidelines as well as Simula’s Code of Ethics and Publication guidelines. Work will be aligned with the EU’s Ethics Guidelines for Trustworthy Artificial Intelligence and the EU AI Act. This research focuses on fundamental methods development in controlled game environments where no gender or security considerations are present. Beyond the project’s lifetime, applications of technology developed in FIND would necessitate a thorough evaluation of potential biases, security assessments for memory systems, ethical frameworks for enhanced reasoning capabilities, and data governance protocols for real-world applications.

1.3 Novelty and ambition

FIND’s novelty lies at the intersection of three established fields in a configuration no prior system has achieved simultaneously:

- **Temporal predictive coding** [14] with **distributed external memory**—not just recurrent state: prior tPC models store context solely in hidden activations; FIND externalises memory into content-addressable HDC slots persistent across thousands of timesteps.
- **Memory-augmented networks** [23] with **biologically-grounded iterative inference**—not single-pass read/write: prior MANNs use a single forward pass to address memory; FIND uses iterative predictive-coding inference to resolve ambiguous queries, yielding a $1.75\times$ recall advantage at $\sigma=0.4$.
- **Hyperdimensional computing** [11] **embedded as the storage format of a gradient-trained system**—not as a standalone symbolic method: prior HDC systems operate outside neural training loops; FIND integrates HDC binding directly into the differentiable memory update, enabling end-to-end learning.

The orchestration mechanisms—anchored in NCA and message-passing with a structured ablation ladder—represent a new class of learning algorithms, grounded in our published work [51, 52], that balances competition and collaboration between specialised modules. FIND’s hypothesis that modular specialisation can emerge naturally from predictive principles challenges conventional wisdom about hand-engineered architectural features; our preliminary results provide the first empirical evidence for this claim, with PM already achieving $3.75\times$ less forgetting than the tPC-alone baseline on controlled MNIST benchmarks.

2 Impact

2.1 Potential for academic impact of the research project

If successful, FIND will deliver innovative outputs addressing fundamental limitations in current AI systems. Our preliminary PM implementation already achieves 98.1% exact recall at 40% input noise and $3.75\times$ less forgetting than the ablated baseline—establishing a quantitative floor on which PAM will build. We will develop the PAM architecture—a novel framework integrating modularized memory systems, active temporal predictive coding, and hyperdimensional computing. This architecture utilizes memory-augmented neural networks for multi-scale information storage and symbolic representation, with orchestration mechanisms that dynamically route information between specialized modules for efficient temporal data processing. PAM functions as a causal world model, enabling understanding of cause-effect relationships beyond statistical correlations and reasoning about interventions and counterfactuals. We will enhance these systems with adaptive learning mechanisms balancing plasticity for new information with stability for knowledge retention. All components will be released as open-source software with comprehensive benchmarks and documented in high-impact publications.

The FIND project is poised to generate significant impacts across multiple stakeholder groups. For AI researchers, our distributed memory mechanisms offer a fundamentally different approach to handling information over extended time periods, potentially resolving context window limitations of current models. The demonstrated effectiveness of modular specialization will shift how researchers design complex AI systems—away from monolithic architectures toward ensembles of specialized modules that dynamically collaborate based on task demands. The synthesis of memory, causal reasoning, and symbolic representations will fundamentally advance how information is stored, processed, and utilized in neural networks. For computational neuroscientists, our modular architecture offers new tools for understanding how specialized brain regions emerge from simple principles—maturing into a unified theory connecting memory, prediction, and action.

All software, datasets, and trained models will be released as open-source projects on GitHub under MIT/Apache 2.0 licenses, with comprehensive documentation following FAIR principles. All trained models will be hosted on Hugging Face under versioned releases, enabling direct comparison and reproducibility across architectures.

2.2 Societal relevance and stakeholder engagement

Beyond academia, FIND addresses needs in three concrete sectors where long-term memory and causal reasoning are bottlenecks. **(1) Healthcare AI:** Clinical decision-support systems must integrate patient histories spanning years, yet current models forget context beyond their context window; PAM’s explicit memory could enable diagnostics that reason across a full patient timeline without re-processing. **(2) Industrial automation and logistics:** Production planning (exemplified by our Factorio benchmark) requires reasoning about causal chains across thousands of steps; manufactur-

ers and logistics operators—including Simula’s existing industry partners in the SURE-AI center and beyond—need agents that accumulate operational knowledge over production runs rather than resetting per shift. **(3) Environmental monitoring:** Climate and ecological agents must track slow-drift processes over months and years; the hierarchical slow/fast memory architecture developed in WP1 (T1.3) is directly applicable to long-range sensor fusion.

To ensure non-academic perspectives shape the research agenda, we will: (i) offer a 6-month knowledge-transfer internship for the postdoc at one industry partner during Year 3; and (ii) submit FIND results to the NORA Tech and SIMULA Innovation programme to assess commercialisation pathways for the PAM memory module. These activities are designed to ensure that the fundamental research outputs of FIND remain grounded in real-world needs throughout the project.

2.3 Measures for communication and exploitation

The communications strategy will be coordinated by Simula; see CV. *The research will be shared with peers* via presentations at NeurIPS, ICLR, AAAI, ICML, COSYNE, CCN, NaiSYS, and the Nordic Neuroscience Conference. High-impact publications will target the Nature and Science portfolio, PNAS, and IEEE/ACM Transactions. Results will be presented to students through the NORA NeuroAI network and the NeuroAI course (Master’s/PhD level) at UiO led by Lepperød, and through the interdisciplinary honors BSc program at UiO directed by Malthe-Sørenssen. Simula’s established communication office will coordinate broader public outreach via press releases and media contacts. *Findings relevant to the three societal sectors* will be shared directly: healthcare AI via the Norwegian Centre for E-health Research; industrial automation via Simula’s SURE-AI partners; and environmental monitoring via the Norwegian Meteorological Institute and NORA applied AI channels.

Data Management Plan (DMP): All PAM software will be released under MIT or Apache 2.0 licenses via GitHub with continuous-integration release cycles tied to project milestones. All publications will be made open-access, with preprints on arXiv or bioRxiv accompanied by code, parameter settings, and supplementary analyses (see the online form of the proposal).

3 Implementation

3.1 Project manager and project group

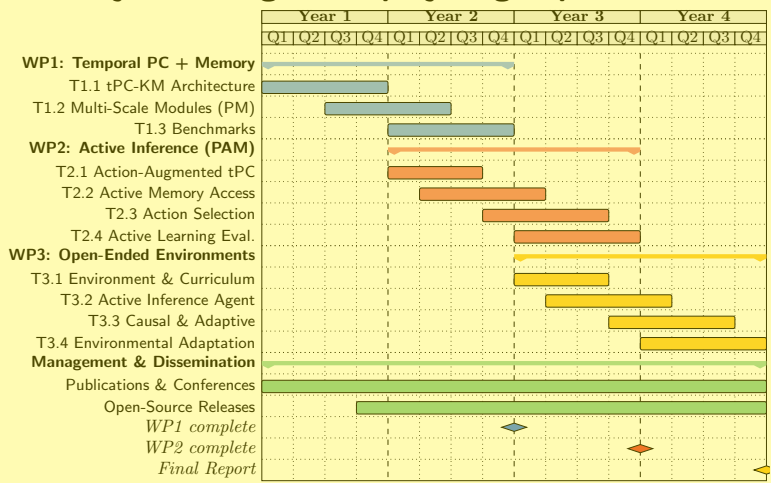


Figure 3: FIND project Gantt chart. WP1 (blue) Years 1–2; WP2 (red) Years 2–3; WP3 (yellow) Years 3–4. Lepperød leads Year 1; postdoc joins Year 2.

maps in spatial navigation, with research on spatial cells and remapping featured in high-impact journals such as NeurIPS, eLife, Science Advances and Nature Communications. Dr. Lepperød’s work on causal inference, particularly through quasi-experimental methods in neuroscience, provides a knowledge bank to the FIND project’s objectives of developing causal world models. His dedication to open science and software development, along with his role in establishing NeuroAI as a prominent research field in Norway, positions him uniquely to drive this innovative interdisciplinary project forward.

The interdisciplinary advisory board comprises collaborators of Dr. Lepperød: **Dr. Michael A. Riegler** (Head of AI, Simula Met) will guide WP3 on advanced AI integration; **Dr. Marc de Kamps** (Professor, University of Leeds) contributes expertise in computational neuroscience and compositional learning for WP2–3; **Dr. Konrad P. Körding** (Professor, University of Pennsylvania) aligns Bayesian decision theory and causal inference with WP1–2; and **Dr. Anders Malthe-Sørenssen** (Professor, University of Oslo) provides interdisciplinary project leadership expertise for WP3 integration.

Dr. Mikkel Elle Lepperød leads the BioAI group at Simula Research Laboratory, where he explores the synergy between neuroscience and artificial intelligence. He holds a master’s degree in applied mathematics and a PhD in neuroscience from the University of Oslo. In AI, his group has developed neural orchestration mechanisms—including neural cellular automata for inter-module communication and one-shot memory transfer—and memory-augmented predictive coding achieving state-of-the-art recall under noise, with results presented at NeurIPS. His interdisciplinary approach has contributed significantly to understanding cognitive

National and international collaboration will support the academic development of the postdoc through research exchange (Internationalisation funds planned from RCN). We have a broad national network as part of the NeuroAI special interest group, which organizes two annual workshops providing a strong national collaborative environment; see CV. Our international collaborators are leading researchers in AI, computational neuroscience, and causality.

3.2 Project organization and management

Management: The project leader, M. Lepperød, is responsible for the day-to-day management and reporting to the RCN. In Year 1, Lepperød will conduct foundational PM implementation and initial benchmarks (T1.1–1.2) while recruiting the postdoctoral researcher; the PD will join at the start of Year 2 and lead WP2 tasks in close collaboration with Lepperød on WP1 completion.

Work Package organization and task allocation: The Gantt chart outlines the work plan, including work packages and tasks. **Research infrastructure:** The FIND project will be hosted at Simula in the Department of Numerical Analysis and Scientific Computing (SCAN) and be part of the BioAI group. As host, Simula provides modern facilities and access to the eX3 infrastructure. FIND will also have access to UiO's extensive computational infrastructure as well as sigma2 Saga and Olivia. WP3 RL experiments in Minecraft and Factorio are estimated to require $\sim 10^5$ GPU-hours in total (based on ~ 800 GPU-hours per DreamerV3-scale training run $\times \sim 100$ experiments across WP2–3); this will be distributed across Sigma2 Saga (project allocation requested) and Simula's eX3 cluster.

Researcher training: The research group (SCAN, BioAI) has extensive experience with supervision and building successful research activities. As part of Simula, FIND will also have access to all the researcher training programs and support offered by Simula Academy, and additional postdoc training will be done in collaboration with the NORA research school.

References

- [1] Y. LeCun. <https://openreview.net/pdf?id=BZ5a1r-kVsf>. 2022.
- [2] Y. Razeghi et al. (2022). arXiv: 2202.07206.
- [3] J. Pasukonis et al. (2022). arXiv: 2210.13383.
- [4] J. Richens et al. (2024). arXiv: 2402.10877.
- [5] L. Wang et al. (2024). DOI: 10.1109/tpami.2024.3367329.
- [6] F. Deng et al. (2023). arXiv: 2307.02064.
- [7] M. Suzuki et al. (2023). DOI: 10.1038/s41583-023-00756-z.
- [8] Y. Wu et al. 2018. arXiv: 1804.01756.
- [9] P. Kanerva. 2003. ISBN: 978-0-262-11132-4.
- [10] A. H. Marblestone et al. (2020). arXiv: 2002.02385.
- [11] P. Kanerva (2009). DOI: 10.1007/s12559-009-9009-8.
- [12] K. Friston (2010). DOI: 10.1038/nrn2787.
- [13] B. Millidge et al. 2022. DOI: 10.24963/ijcai.2022/774.
- [14] B. Millidge et al. (2024). DOI: 10.1371/journal.pcbi.1011183.
- [15] R. P. N. Rao et al. (2023). ISSN: 1530-888X. DOI: 10.1162/neco_a_01627.
- [16] P. Lewis et al. (2020).
- [17] C. Zhang et al. (2023). arXiv: 2309.01390.
- [18] M. Laskin et al. (2022). arXiv: 2210.14215.
- [19] Bulbagarden. <https://bulbagarden.net/threads/305796/>. 2025.
- [20] M. Gemici et al. (2017). arXiv: 1702.04649.
- [21] C. R. Gallistel et al. 2011. ISBN: 978-1-405-12288-7.
- [22] A. Graves et al. (2014). arXiv: 1410.5401.
- [23] A. Graves et al. (2016). DOI: 10.1038/nature20101.
- [24] J. Ramapuram et al. (2021). arXiv: 2103.03905.
- [25] Y. Wu et al. (2018). arXiv: 1811.09556.
- [26] N. Jones (2025). DOI: 10.1038/d41586-025-00649-4.
- [27] D. Ma et al. (2022). arXiv: 2207.12932.
- [28] B. Aragam et al. 2024. DOI: 10.52202/079017-3211.
- [29] B. Scholkopf et al. (2021). DOI: 10.1109/jproc.2021.3058954.
- [30] K. Drozdov et al. (2024). arXiv: 2412.10925.
- [31] G. Zhou et al. (2024). arXiv: 2411.04983.
- [32] Z. Hu et al. (2024). arXiv: 2402.07465.
- [33] T. Ni et al. (2023). arXiv: 2307.03864.
- [34] T. Hafting et al. (2005). DOI: 10.1038/nature03721.
- [35] J. O'Keefe (1976). DOI: 10.1016/0006-8993(76)90358-7.
- [36] J. C. Whittington et al. (2020). DOI: 10.1016/j.cell.2020.10.024.
- [37] A. Banino et al. (2018). DOI: 10.1038/s41586-018-0102-6.
- [38] V. Schøyen et al. (2022). ISSN: 1556-5068. DOI: 10.2139/ssrn.4309105.
- [39] M. B. Pettersen et al. (2025). DOI: 10.7554/elife.99302.
- [40] M. Lepperød et al. 2024. DOI: 10.52202/079017-0008.
- [41] V. S. Schøyen et al. (2025). ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1012804.
- [42] J. Gornet et al. (2024). DOI: 10.1038/s42256-024-00863-1.
- [43] M. Tang et al. (2023). DOI: 10.1371/journal.pcbi.1010719.
- [44] E. M. Russek et al. (2017). DOI: 10.1371/journal.pcbi.1005768.
- [45] K. L. Stachenfeld et al. (2017). DOI: 10.1038/nn.4650.
- [46] J. Nassar et al. (2022).
- [47] C. V. Nguyen et al. (2017). arXiv: 1710.10628.
- [48] L. C. Melo et al. (2024). arXiv: 2410.07812.
- [49] M. Wasiluk et al. <https://openreview.net/pdf?id=2LhCPowI6i>. 2023. arXiv: 2309.01895.
- [50] A. Singh et al. (2025). arXiv: 2503.04728.
- [51] M.-K. Kvalsund et al. (2026). DOI: 10.1016/j.neunet.2026.108798.
- [52] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.831.
- [53] E. Guichard et al. 2025. DOI: 10.1162/isa1.a.833.
- [54] M. A. Alsakkal et al. (2025). arXiv: 2501.18799.
- [55] J. C. R. Whittington et al. (2021). arXiv: 2112.04035.
- [56] M. Lewis et al. (2019). DOI: 10.3389/fncir.2019.00022.
- [57] M. Pettersen et al. (2025). DOI: 10.1101/2025.09.25.678504. eprint: <https://www.biorxiv.org/content/early/2025/09/25/2025.09.25.678504.full.pdf>.
- [58] B. M. Lake et al. (2015). DOI: 10.1126/science.aab3050.
- [59] M. Johnson et al. 2016. DOI: 10.24963/ijcai.2016/643.
- [60] S. Dziadzio et al. (2023). arXiv: 2312.16731.
- [61] K. He et al. 2016. DOI: 10.1109/cvpr.2016.90.
- [62] J. Chung et al. (2015). arXiv: 1506.02216.
- [63] S. Hochreiter et al. (1997). DOI: 10.1162/neco.1997.9.8.1735.
- [64] A. Gu et al. (2021). arXiv: 2111.00396.
- [65] K. Park et al. (2023). arXiv: 2311.03658.
- [66] A. Nichol et al. (2018). arXiv: 1803.02999.
- [67] J. Schulman et al. (2017). arXiv: 1707.06347.
- [68] A. Mete et al. 2024. arXiv: 2402.00158.
- [69] T. Yu et al. (2019). arXiv: 1910.10897.
- [70] N. Wagener et al. (2022). arXiv: 2208.07363.
- [71] W. Hill et al. (2023). arXiv: 2312.12891.
- [72] N. Kant (2025). arXiv: 2502.01492.
- [73] R. Portelas et al. (2020). arXiv: 2003.04664.
- [74] A. Aubret et al. (2023). DOI: 10.3390/e25020327.
- [75] M. Andrychowicz et al. 2017. arXiv: 1707.01495.
- [76] V. Mnih et al. (2015). DOI: 10.1038/nature14236.
- [77] Y. Burda et al. (2018). arXiv: 1808.04355.
- [78] D. Hafner et al. (2018). arXiv: 1811.04551.
- [79] L. Espeholt et al. (2019). arXiv: 1910.06591.
- [80] J. Schrittwieser et al. (2019). arXiv: 1911.08265.